

# Spectral Radius Minimization for Signed Squares of Cycles

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## Abstract

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## 1 Introduction

## 2 Switching Coordinates and the Reference Phase

### 2.1 Edge signings, switching, and Hamilton gauge

### 2.2 Flux words, holonomy, and the attaining candidate

### 2.3 The squared local operator

### 2.4 The period-eight reference phase

## 3 Gaps, Charges, and Translation Sectors

### 3.1 Cyclic gaps and excess charge

### 3.2 Translation sectors

### 3.3 Composition and legal cyclic closure

## 4 The Elementary Six-Gap Phase Slip

### 4.1 The bilateral interface and its essential spectrum

### 4.2 Algebraic candidate and physical matching

### 4.3 The global G6 edge

### 4.4 Rank-two symmetry

## 5 Optimality Among Single-Gap Interfaces

### 5.1 Canonical single-gap operators

### 5.2 Exact finite-gap comparisons

### 5.3 Uniform control of the large gaps

### 5.4 Reference and single-gap hierarchy

## 6 Phase Slips on Finite Rings

### 6.1 Legal residue constructions

### 6.2 Identification of local patches

### 6.3 Discrete IMS localization

### 6.4 Residue upper bounds and the analytic tail

### 6.5 Structural refinement for separated interfaces

## 7 Finite Completion of the Classification

### 7.1 Exact-computation protocol

### 7.2 Orders 8 through 32

### 7.3 Orders 34 through 46

### 7.4 The exact bridge from 48 to 238

### 7.5 Exhaustive synthesis

## 8 Concluding Remarks

## Data and Code Availability

### A Exact Spectral Certification of the Reference Phase and G6

### B Exact Finite Classification and Completeness

## References

- [1] Yonatan Bilu and Nathan Linial. Lifts, discrepancy and nearly optimal spectral gap. *Combinatorica*, 26(5):495–519, 2006.